

Predicting Final-Hour SPX Price Movements for Short-Dated Options Trading Using Intraday Market Features

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1 Overview

Often in the last hour of trading, financial markets frequently see increased volatility, especially in the S&P 500 index (SPX), where Zero-days-to-expiration (0DTE) options are extremely vulnerable to sudden price changes. Although it is challenging to consistently be able to predict these short-term movements, they can present significant trading opportunities.

This dissertation explores the possibility of using intraday market behavior patterns to forecast notable price changes during the last trading hour. Machine Learning (ML) models will be created to determine whether the market is likely to move significantly upward, downward, or remain relatively stable prior to market close by analyzing historical price and volatility data.

Additionally, the study will also look at whether these forecasts can enhance decision-making in a streamlined 0DTE options trading approach. The objective is to close the gap between statistical modeling and real-world trading applications.

2 Aim

To develop and evaluate ML models that predict the direction and magnitude of SPX price movements during the final trading hour, and to assess their usefulness in guiding short-dated options trading strategies.

3 Research Questions

This study aims to investigate the predictive power of intraday market features on end-of-day price movements in the SPX, as well as their applicability in options trading strategies. The research is guided by the following questions:

1. **RQ1:** Can intraday market features be used to predict the direction of SPX price movements during the final hour of trading?
2. **RQ2:** To what extent can intraday features predict the magnitude of SPX price movements during the final hour of trading?
3. **RQ3:** Which intraday features (e.g., momentum, volatility, and price positioning) are most important in predicting large end-of-day movements?
4. **RQ4:** Can SPX price movements in the final hour be predicted with sufficient accuracy to be economically meaningful for trading applications?
5. **RQ5:** Can these predictions be used to construct a profitable end-of-day options trading strategy after accounting for transaction costs and realistic execution constraints?

4 Motivation and Research Significance

Many traders and researchers believe that 0DTE options represent one of the best ways to speculate, hedge or mitigate potential risks associated with daily trading. Many recent studies on markets show that today 0DTE options are responsible for a majority of the SPX options that are traded; this may be the reason why it has become such an important factor for researchers studying day-to-day price behavior TabbFORUM (2026), Cboe Global Markets (2026).

Since 0DTE options' prices can fluctuate quickly in response to even small movements in the underlying index in the last hour of trading (due to rapid theta decay, high levels of gamma exposure or acceleration, and the fast moving delta) movements in the SPX over the last hour of trading (3:00 - 4:00 pm) can result in large percentage changes in the value of near-the-money and/or slightly out-of-the-money option prices. Since the final hour of trading offers a potentially greater risk/reward environment than other times throughout the trading day, examining short-term directional moves in price can provide researchers with new opportunities.

We designed the current study with the belief that SPX price movement at the end of the trading session may not be entirely randomly distributed. Research studies have shown that intraday return distributions vary throughout the trading session and may be influenced by such variables as volume. Kawaller et al. (1990) laid some of the early groundwork for understanding frequency relationships among volatilities of the SPX Index and SPX Futures. Later studies have expanded upon their findings, illustrating that ML models which utilize realized volatility and high-frequency financial data features can offer superior predictions for returns and volatility Zhang et al. (2024), Ferreira & Medeiros (2021).

Our central hypothesis is that features derived from intraday market behavior contain sufficient information to improve prediction of future direction and magnitude of SPX price movements during the final hour compared to simple

baseline models and that these results could support a more systematic approach to making decisions related to same-day 0-DTE option trading.

5 Literature Review

This literature review discusses papers that relate to predicting final hour SPX price movements and it will be organized around three major themes that assist in answering the research questions: intraday return and volatility predictability, high frequency financial forecasting using ML techniques, option-implied information and 0DTE market structure, and the gaps of previous studies. Instead of presenting each study individually, the literature review will address how the literature supports the design of this dissertation.

Table 1: Summary of key literature and relevance to the dissertation

Theme	Study	Main contribution	Relevance to this dissertation
Intraday volatility predictability	Andersen & Bollerslev (1998)	Demonstrates that volatility models can provide useful forecasts when evaluated appropriately, especially when using higher-frequency information.	Supports the assumption that intraday market behaviour contains structure that may be useful for forecasting SPX movement.
Intraday volatility predictability	Poon & Granger (2003)	Reviews volatility forecasting literature and shows that forecast performance depends on data frequency, model choice, forecast horizon and evaluation method.	Justifies evaluating the final-hour prediction problem using a horizon-specific framework rather than relying only on general daily volatility evidence.
Intraday market behaviour	Kawaller et al. (1990)	Documents intraday relationships between SPX futures volatility, SPX index volatility and trading volume.	Supports the use of OHLCV-derived predictors such as realised volatility, volume behaviour, intraday range and time-specific price movement.
ML and high-frequency forecasting	Ferreira & Medeiros (2021)	Shows that ML models can improve intraday return forecasting using minute-level market data, lagged returns, realised-volatility measures and VIX information.	Supports the use of non-linear supervised learning models and volatility-based predictors for short-horizon SPX forecasting.
ML and realised volatility	Zhang et al. (2024)	Compares HAR-type models with ML models such as penalised regression, XGBoost, MLP and LSTM for realised-volatility forecasting.	Supports testing multiple model types and accounting for intraday and time-of-day effects rather than relying on a single linear benchmark.
ML and market microstructure	Rahimikia & Poon (2026)	Finds that ML models using high-frequency limit-order-book and news-related variables can outperform traditional realised-volatility benchmarks.	Highlights the value of high-frequency features, while also showing a limitation of this dissertation because only OHLCV-derived features are available.

Theme	Study	Main contribution	Relevance to this dissertation
Option-implied information	Fan et al. (2026)	Shows that option-implied rough volatility information can improve realised-volatility forecasts beyond traditional HAR-RV/VIX-based models.	Supports the idea that options-market information contains useful predictive content, although this dissertation focuses mainly on observable intraday SPX features.
0DTE trading performance	Vilkov (2023)	Examines realised payoffs from 0DTE SPX option strategies and shows that returns can be skewed, path-dependent and sensitive to liquidity and timing.	Justifies evaluating predictions through an economic trading lens rather than relying only on classification or regression accuracy.
0DTE gamma risk	Dim et al. (2024)	Studies 0DTE trading, gamma exposure and volatility propagation, documenting links between market-maker positioning and subsequent realised volatility.	Supports the importance of short-horizon market conditions and motivates using volatility, momentum and range-based proxies when direct gamma data are unavailable.
0DTE market impact	Vasquez et al. (2025)	Documents relationships between lagged gamma-related positioning and subsequent realised volatility in the underlying market.	Provides further support for examining whether intraday market conditions before the close contain information about later SPX movement.
Jump risk and 0DTE options	Bozovic (2025)	Studies intraday jumps in SPXW 0DTE options and their implications for pricing and hedging same-day options.	Supports predicting both direction and magnitude because sudden final-hour moves can strongly affect 0DTE option outcomes.
Market relevance and practitioner evidence	Cboe Global Markets (2023, 2026), Xu (2025), TabbFORUM (2026)	Documents the growth and market relevance of SPX 0DTE options.	Motivates the dissertation topic by showing that 0DTE options are now a major part of SPX options trading rather than a niche market.
Practitioner risk evidence	Numerix & Risk.net (2024)	Discusses practical risk issues linked to zero-day options, including intraday hedging pressure and short-dated gamma exposure.	Provides practitioner support for considering execution costs, hedging pressure and market frictions when evaluating a 0DTE trading strategy.

Intraday Market Behaviour and Volatility Predictability

A primary assumption of this dissertation is that intraday market behavior includes identifiable structure. Without some degree of structure within short-horizon SPX movements, there is no logical reason for utilizing ML to predict the final trading hour. The extant financial literature has demonstrated that while volatility persistence is a well established concept Andersen & Bollerslev (1998), the extent to which volatility is predictable depends upon the chosen data, model, forecast horizon and/or evaluation methodology Poon & Granger (2003). Further, Andersen & Bollerslev (1998) provided empirical evidence showing that high-frequency data is capable of revealing volatility patterns that do not appear

at lower frequencies.

Previous research has also shown that intraday volatility is associated with market activity and time-of-day effects in the context of SPX related markets. Kawaller et al. (1990) documented systematic relationships between futures volatility, index volatility and trading volume. While this study is not recent, its findings remain relevant as they establish that SPX intraday behavior is not independent of trading activity. Thus, the study supports the inclusion of features such as realized volatility, volume behavior, intraday range and time based price movement in the dissertation’s model.

These studies together help us justify the methodology chosen for this dissertation and they also support the notion that variables observed prior to 3:00 p.m. contain information about subsequent price behavior. However, these studies have also illustrated that volatility forecasting ability does not necessarily imply either directional predictability or trading profitability. Thus, this dissertation builds upon the foundation established by these studies by examining whether intraday features can be used to predict both the direction and magnitude of SPX movement during the final hour.

ML and High-Frequency Financial Forecasting

The relationship between intraday market and final hour SPX movement is non-linear and complex. Therefore, it is unlikely that all of the potential features identified above (e.g. lagged returns, realized volatility, volume acceleration, Volume Weighted Average Price (VWAP) distance, intraday high-low positioning, etc.) will act independently. ML methods are particularly suitable for modeling non-linear relationships between features.

Ferreira & Medeiros (2021) found that ML models improved intraday return forecasting when compared to linear models when applied to minute-level market data they used lagged returns, realized-volatility measures and VIX information to find relative performance among different approaches (i.e. Random Forests vs. LSTM models). Their results provide support for employing both price-based and volatility-based predictors in this dissertation. Additionally, their study showed that VIX-related information can be beneficial for incorporating implied volatility. However, their study focused on minute-by-minute return forecasting as opposed to a clear final-hour SPX classification problem.

Zhang et al. (2024) compared traditional HAR-type (Heterogeneous Autoregressive models) models with ML models (penalized regression, XGBoost, multi-layer perceptron, and LSTM models) for realizing volatility forecasting. They also found that ML approaches generally outperformed traditional HAR-type models when considering intraday similarity. Similarly, Rahimikia & Poon (2026) reported that ML models using higher quality high frequency information (limit-order-book and news-related variables) were able to outperform traditional realized-volatility benchmarks over a substantial portion of the out-of-sample period. They also provided explainability results indicating that bid/ask/mid-price variables were important predictors.

Altogether, this literature supports the application of supervised ML models to intraday financial features. Also, above literature suggests that one should evaluate multiple model types rather than solely rely on a single linear benchmark. Notably, however, studies typically employ rich microstructure data (limit-order book, bid-ask spread, and news variables) whereas this dissertation employs only OHLCV-derived features and technical indicators due to constraints in accessing higher quality and often expensive data.

Option Implied Information and 0DTE Market Structure

Vilkov (2023) documented that when researching the payoffs from multiple 0DTE SPX options trading strategies, the returns generated by each strategy were extremely skewed towards negative returns, highly dependent upon the path taken during implementation, and highly dependent upon liquidity and timing of trade. In general, this demonstrates that regardless of the accuracy of statistical predictions, those predictions should be evaluated through an economic lens. For example, while a model might correctly predict direction, it would fail as a viable trading strategy if the size of the final hour move was too small; if the premiums paid for trading were too high; or if bid/ask spread and other transaction costs reduced profitability.

Dim et al. (2024) and Vasquez et al. (2025) published additional research documenting how important short term market conditions are to 0DTE market makers' performance when exploring 0DTE trading, gamma exposure (GEX), and volatility. The authors found statistically significant relationships between market maker gamma exposure and subsequent realized volatility as well as intraday market behavior. Both papers report statistically significant associations between observed 0DTE activity and observed changes to the underlying market. Because obtaining direct gamma exposure data is often difficult, this dissertation uses OHLCV-derived proxies such as realized volatility, range expansion, momentum, and price positioning to represent state information regarding the underlying market.

Another key factor to consider for 0DTE options is "jump-risk" due to potential loss from abrupt price movements near expiration, which can negatively affect same-day options traders. Bozovic (2025) investigated intraday jumps in SPXW 0DTE options and documented implications of discontinuous price movements on pricing and hedging same-day options. The results confirm that the final hour is an economically meaningful period to study since rapid large movements at the end of a session can greatly effect the success of same-day options.

In summary, there exists a substantial amount of literature suggesting that 0DTE options act differently than longer-term option contracts. Short-term options are heavily affected by several variables including: the proximity to expiration (i.e. time-to-expiration), GEX, jump-risk or gaps, and execution costs/trade-friction. The combined effects of all of these variables create a unique set of characteristics that determine the value of short-term options.

Market relevance and practitioner evidence

Although peer-reviewed research provides the bulk of academic reference materials to this topic, practitioner and trade reports help us in establishing relevance of topic in current age. Cboe Global Markets (2023, 2026), Xu (2025) reports clearly indicate the explosive growth of SPX 0DTE (same day expiration) trades and demonstrate that same day expiration options now constitute a significant portion of all SPX option volume. Also, Numerix & Risk.net (2024) have discussed the risks associated with zero-day options, such as intra-day hedging pressures and short-dated gamma exposure.

Limitations of existing research and research gap

While there is considerable evidence within the literature supporting each of the three areas addressed here, there are significant gaps remaining that we plan to answer in the dissertation. Firstly, while numerous intraday volatility studies have shown that high frequency market behavior has some sort of structure, few have examined how to utilize this knowledge to accurately predict SPX price movement at a specific time. Secondly, the majority of ML studies evaluating predictive capabilities of various models utilizing intraday financial data use statistical metrics such as mean absolute percentage error, directional accuracy etc. as a measure of performance; however, they rarely provide a link to a viable trading strategy. Finally, while option studies examining zero days to expiration (0DTE) typically examine the distribution of payoffs but none examine if OHLCV derived intraday features can actually predict the direction and magnitude of SPX price changes prior to closing.

The objective of this dissertation is to address each of these gaps by combining all three areas into one focused empirical study. This dissertation utilizes intraday OHLCV-derived features, technical indicators, and volatility measures to predict SPX price changes occurring within the last hour of trading and also checks them for a profitable strategy.

6 Exploration of Data and Methods

Artefact Development and Analytical Workflow

The artefact workflow in Figure 1, represents the planned workflow of artefacts for this dissertation. The artefact workflow provides a visualisation of how raw intraday data will be processed from cleaning, feature engineering, target creation, exploratory analysis, model development and finally evaluate the suitability of the trade for final trading evaluation. The artefact workflow is an important part of the overall technical pipeline as it demonstrates that the technical pipeline is connected to the research question.

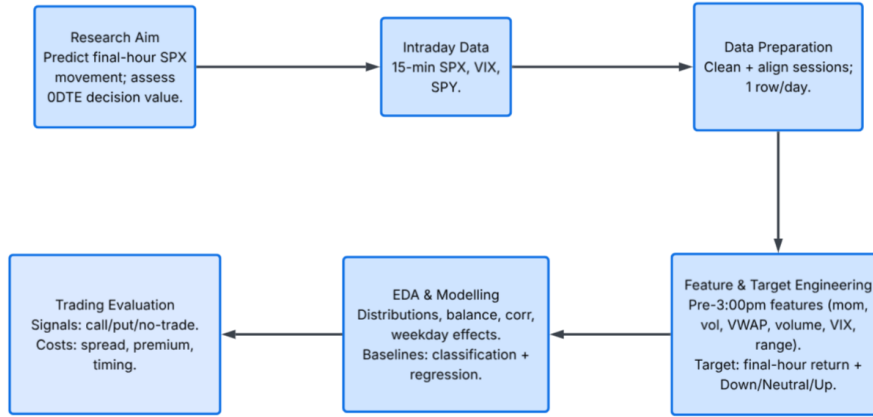


Figure 1: Analytical Workflow

Data Collection and Data Description

Yahoo Finance was used to collect data at 15 minute intervals. This was chosen due to its accessibility and suitability for developing a small-scale proof of concept based on an eventual larger history-based dataset for the full dissertation.

GSPC (SPX) and VIX were included for their relationship with SPX price movements. SPY was also included for its use as a proxy for tradeability, liquidity, and VWAP related characteristics. SPY was selected since there is no direct measure of trading volume in the SPX.

Every valid trading day was transformed into one single modeling observation and features that could be determined prior to the 3 pm decision point were generated to avoid look-a-head bias. The target variable was defined as the SPX return from approximately 3 pm until the end of the trading day. Several of the engineered variables; return, momentum, volatility, ATR, intraday-range-

position, VWAP-distance, SPY-volume, VIX-movement and day-of-the-week-features were developed.

This dataset is being utilized as a means of establishing a proof of concept for validating that all elements of the data pipeline and feature engineering process and model architecture are functioning as intended prior to increasing the scope of this analysis.

Ethical Considerations

An ethical approval application was completed for this dissertation as part of the DkIT research ethics process. The data used in this project is publicly accessible market data and does not contain personal or identifiable information. As a result, there are no direct risks to participants, and informed consent procedures are not applicable. The research data and project files are stored privately using DkIT OneDrive and Github with password and multi-factor authentication protection. Access is restricted to the researcher and supervision team.

Data Cleaning and Feature Engineering

Before extracting features from the intraday data, the raw data was reviewed for potential errors in the form of missing values, duplicate timestamp entries, summary statistics (e.g., mean, median), and incomplete trading days. It was calculated that a full trading day would include 26 fifteen-minute time intervals for SPX, SPY, and VIX. Reviewing the intraday data for such irregularities is necessary since there may be numerous gaps due to holidays, early closures or other download-related problems. For this study only full trading days were reviewed.

In order to remove any trading days which contained missing or invalid data from the modeling dataset, those trading days which did not contain the complete 26 fifteen-minute bars for each tickers were removed. An example of an irregular trading day occurred on May 25th where the data for both SPX and SPY was unavailable but VIX data was still available. The removal of such cases ensured that all features extracted were based upon consistent daily observation periods.

Once the data was processed to clean it of any uncomplete entries, it was transformed into a single row per trading day. All variables were created using only the relevant historical data available prior to 3:00 P.M. The main target variable was the final-hour SPX return which was defined as the difference between the closing value of SPX and its 3:00 P.M. price. Additionally, a classification label for each trading day was developed to categorize each trading day as either "Down", "Neutral," or "Up" to help classify direction.

All engineered features captured various aspects of the market including momentum, volatility, position within the range, volume of SPY and behavior of VWAP for SPY and VIX, and day-of-week effects. The inclusion of these

Table 2: Summary of Data Cleaning Checks

Check	Purpose	Outcome
Missing values	To identify gaps in the downloaded intraday data.	Incomplete days were removed before modelling.
Duplicate timestamps	To ensure that no 15-minute bar was repeated.	Timestamps were checked before feature creation.
Full trading day check	To confirm that each valid day contained 26 15-minute bars for SPX, SPY and VIX.	Only complete trading days were kept.
Market holiday and early-close checks	To identify non-standard trading sessions or inconsistent downloaded data.	Non-standard days, such as May 25th, were excluded.
Post-feature missing values	To confirm that the final engineered dataset was suitable for EDA and modelling.	The final modelling dataset was cleaned and ready for analysis.

features was intended to help capture the overall condition of the markets immediately prior to the final hour and to determine if knowledge of this condition provided any useful insight into how SPX will move during the final hour.

Upon completion of feature extraction, the final modeling dataset was once again evaluated for any missing values to verify that it was prepared for exploratory analysis and model building.

Data Visualisation and Preliminary Analysis

First we wanted to check how the final-hour SPX returns are distributed and as seen in figure 2, there seems to be a large number of final-hour returns which are around zero, but very few have higher (positive) and lower (negative) returns than what would normally be seen at this time. This has significant implications for the dissertation, suggesting that many final-hour periods will be unsuitable for 0DTE trading, even if the model is able to make correct predictions about the direction of price movement.

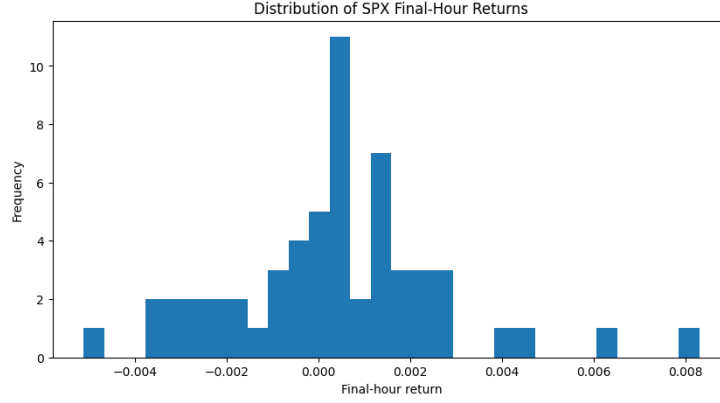


Figure 2: Distribution of SPX Final Hour Return

However, there were also some significantly larger final-hour movements present in the data set. On 7 Apr 2026, the largest upward price movement occurred with the SPX moving from 6563.41 to 6617.92 representing a price increase of \$54.51. The largest downward price movement occurred on 23 Mar 2026 when the SPX moved from 6615.60 to 6581.76 representing a price decrease of \$33.84. These two examples illustrate why the last hour is so important for 0DTE options; while percentage changes in the underlying index may be small, they result in significant point movements.

To classify all final-hour returns appropriately into one of three categories: Down, Neutral and Up, we needed to establish appropriate threshold values. Since the data were drawn randomly from a normal distribution, we used the 33rd and 67th percentiles to establish our upper and lower bounds. We defined Down as all returns less than -0.010%, Up as those greater than 0.120% and Neutral as all returns within the range -0.010% to +0.120%. This resulted in an equally balanced classification problem with 19 observations falling into each category. An advantage of having this equally balanced target variable is that it provides a good way to test early baseline models without allowing them to learn to always predict the most common category. It is essential to note that these thresholds are only intended as exploratory tools and early model evaluation metric. In later stages when developing our dissertation's final versions of the models, we will need to revise new thresholds based upon economic considerations rather than statistical ones.

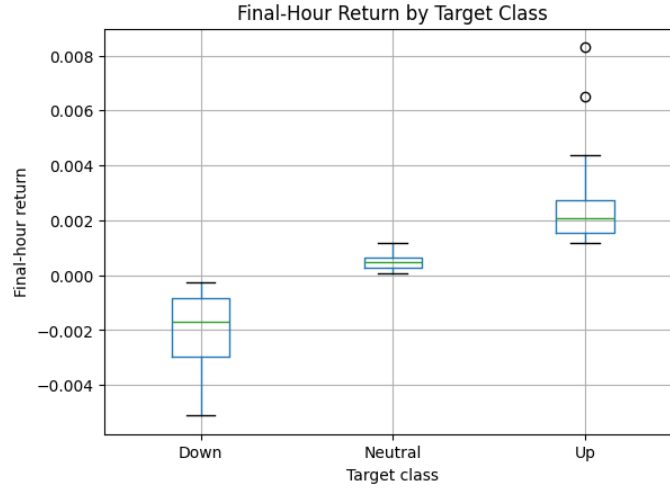


Figure 3: Final Hour Return based on class

In summary, although our preliminary exploratory data analysis shows that the problem of predicting the direction of stock prices during the final hour should not solely be viewed as a directional forecasting issue. Rather, the primary challenge for successful trading will involve identifying when sufficient market movement occurs to produce viable trading opportunities.

Analysis of Day of Week

The initial study's first stage was able to find some differences on different days of the week. However, because there we are currently using a small and free dataset, we need to use caution with the interpretation of these results as they may change later. For example, Friday showed a slightly negative average return for the last hour (-0.000596), whereas Tuesday and Thursday had substantially larger positive average returns (.000805 and .000770) respectively. Also, the class distribution indicated that Thursdays had more than double the number of upward return value classes as compared to downward, whereas the opposite occurred on Fridays (more downward). It suggests that day of the week should be included in all future modeling data sets as an additional control variable.

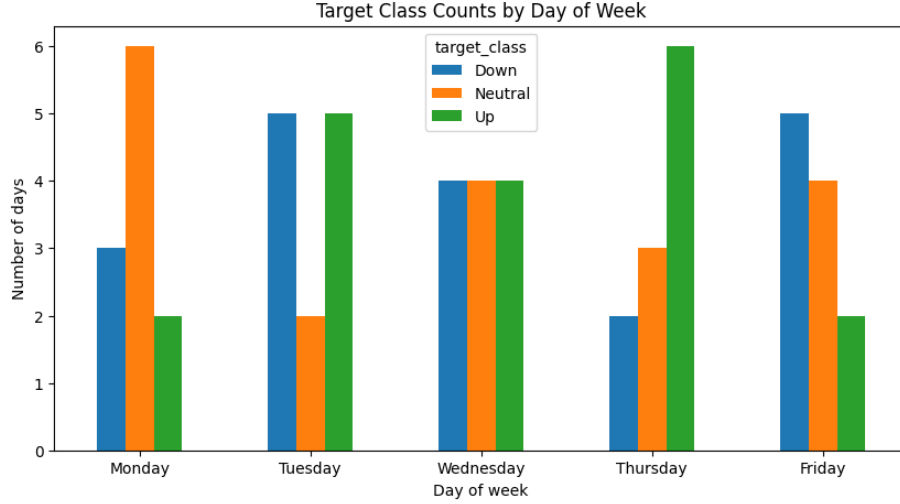


Figure 4: Analysis of Day of Week

Feature based Correlation Analysis

A Feature-Target Correlation Analysis was done to check whether or not any of the Engineered Variables would exhibit a strong linear relationship with Final Hour Return. The purpose of this initial screening process was to help reduce the number of potential input variables, since there are very few observations in the current dataset and many Financial Relationships are Nonlinear.

Positive correlations to the final hour returns were identified for the 60-minute prior return of VIX, with a correlation coefficient value of 0.261. As such, it appears that changes in implied Volatility over the last sixty minutes may contain information related to how SPX performs into closing time. Positive correlations to the final hour returns were also identified for Distance from the Day Low, SPY's position relative to its VWAP, Realised Volatility and Average True Range (ATR). Thus, these features suggest that volatility conditions, Intraday Price Positioning and Market-State Variables are potentially relevant for predicting performance during the final hour.

Table 3: Selected correlations with final-hour SPX return

Feature	Correlation
VIX return last 60 minutes	0.261
Range expansion ratio	0.140
Distance from day low	0.130
SPY distance from VWAP	0.129
Realised volatility last 60 minutes	0.118
ATR last 60 minutes	0.117
Return last 15 minutes	-0.166
Momentum acceleration	-0.213
Return last 60 minutes	-0.239
Return last 30 minutes	-0.282

The largest negative relationships were seen with recent SPX momentum variables. There was a correlation of -0.282 for returns over the past thirty minutes; -0.239 for returns over the last sixty minutes; and -0.213 for the momentum acceleration. These findings indicate possible short-term reversal effects in this initial sample. In particular, it appears there has been greater price movement prior to 3:00 p.m., and then less or counter-movement in the final hour.

While these correlations are weak enough to prevent drawing conclusions based solely on them, they do represent useful beginning evidence supporting the usefulness of VIX movement, recent SPX momentum, range behavior, VWAP distance and volatility measures within an initial modeling framework. They also support further investigation using non-linear ML models later in this dissertation.

Model Evaluation

In addition to evaluating the overall quality of fit for the models I wanted to assess them in terms of their ability to correctly classify each event into one of three categories Down, Neutral, or Up as well as their ability to measure the magnitude of each event. To accomplish this, I will utilize two sets of evaluation criteria. The first set will be directed towards classification models and will consist of accuracy, precision, recall, f1 score and a confusion matrix for each category. I will pay special attention to models' ability to find breakouts in both upward and downward directions. Breakout sessions represent the highest impact opportunities for 0DTE trading.

The second set of evaluation criteria will be applied to regression models and will focus on measuring the errors made by those models when estimating event magnitudes. Errors in estimation will be evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

Additionally, I plan on performing feature importance analyses for my tree-based models. These analyses will tell me which intraday variables contributed

most heavily to my models' final hour predictions.

Where possible, I will use walk-forward validation techniques in order to minimize over-fitting during my model development process and provide a better representation of real-time forecasting conditions.

Proposed Future Analysis

As outlined in Section 2, the immediate next phase of the dissertation will take the interim artefact from exploratory analysis and transform it into predictive modeling and trading evaluations. First, I will need to increase the size of the database of historical intraday prices to allow me to evaluate my models under different market conditions.

I am doing this to avoid having the results from my interim analysis based on a too-short period. It was determined that the best way to do this was to use a variety of supervised ML algorithms. I will start with a multinomial logistic regression model as a linear, interpretable benchmark. In order to compare all other models against something that has been proven to work before, I will use a multinomial logistic regression model. After this, I will develop tree-based models (Random Forest and/or Gradient Boosting or XGBoost) since they are well-suited for finding complex interactions between the many intraday momentum and volatility and volume related variables (VWAP and VIX).

All of the models will be either classification models or regression models depending upon their specific application. Since a 0DTE options position decision relies on knowing where you expect your stock price to go (up/down), as well as whether the expected move is big enough to be considered significant; both types of models will need to be developed. Depending on whether I have sufficient computing resources and access to historical data; I will also try developing a long-short term memory (LSTM) type model. This model will allow me to see if using the sequential nature of intraday bars will improve my predictive accuracy.

Since the financial world has temporal ordering (i.e. prior events affect later ones); it is very important that I use a chronological division of my database into training sets, validation sets and out-of-sample testing sets. If I were to randomly divide my database into training sets and testing sets; it is possible that I could get leakage from previous events affecting my ability to accurately forecast subsequent events.

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URL: <https://doi.org/10.1093/jjfinec/nbad005>

A Generative AI Usage Declaration

Generative Artificial Intelligence has been used to assist with the artefacts developed in this study by providing assistance throughout all stages of artefact development. These tools assisted the researcher in developing a workflow visualization of their planned approach, designing a flow chart for planning how they would execute their plan, and assisting the researcher in debugging and streamlining the programming associated with their analytical pipeline. AI was also used to help with Time Management

A.I. was NOT used as an alternative for my own analysis, interpret the findings, or make any independent academic judgments regarding the project. ALL project related decisions made throughout the course of this study remained my responsibility . For example; what data to select, which features to engineer from that selected data, how to design the models that will be used to analyze the data, how to interpret the results of those analyses, and how to create the final document detailing the results of this study. ANY generated output produced using an A.I.-assisted tool was reviewed, edited and validated prior to its inclusion in either the artefact itself OR as supporting documentation for this study.

The primary purpose of utilizing generative A.I. in this study was to improve upon the efficiency of various support-related processes. Examples include improving upon the overall organization of the coding structure, identifying potential coding errors, creating initial drafts of workflow descriptions, and providing additional clarity to how the artefact development process occurred. NO private/confidential/personal/participant identifiable information was inputted into these A.I. tools. This study utilizes secondary publicly available financial data and does not contain human subjects.

Documentation outlining each prompt utilized when interacting with an A.I. tool during the course of this study and documentation illustrating all AI-assisted interactions have been maintained as part of this study’s documentation in accordance with the requirements provided in the guidelines assessing this project.

A.1 Generative AI Prompt Log

The following prompts represent the types of generative AI support used during artefact development, coding troubleshooting, workflow planning, and project time management. AI-assisted support was used as a development aid only, and all outputs were reviewed, edited, and validated by the researcher.

- How can I structure a reproducible data analytics artefact for predicting final-hour SPX price movements?
- Suggest a clear project workflow for an intraday financial forecasting pipeline.

- How should I organise my Jupyter notebook sections for data collection, cleaning, feature engineering, EDA, and modelling?
- Help design a workflow diagram for a project using SPX, VIX, and SPY intraday market data.
- Suggest a compact flowchart structure that can fit into an interim dissertation report.
- Help create a Lucidchart prompt for visualising the artefact workflow.
- How can I explain the role of SPX, VIX, and SPY in an intraday prediction pipeline?
- Help troubleshoot Python code for downloading 15-minute intraday data from Yahoo Finance.
- How do I check for missing values and duplicate timestamps in intraday financial data?
- Help debug issues caused by incomplete trading days or missing 15-minute bars.
- How can I convert intraday bars into one row per trading day for ML?
- Explain how to avoid look-ahead bias when creating features before a 3:00 p.m. decision point.
- Help create a target variable for final-hour SPX return from 3:00 p.m. to market close.
- How can I convert a continuous final-hour return into Down, Neutral, and Up classes?
- Explain the advantages and limitations of using percentile-based class thresholds.
- Suggest possible alternatives to percentile thresholds for economically meaningful trading signals.
- Help create momentum features such as recent 15-minute, 30-minute, and 60-minute returns.
- Help create realised volatility and ATR-based features from intraday OHLC data.
- How do I calculate distance from the day's high, low, open, and VWAP?
- Explain why VWAP distance and volume acceleration may be useful for intraday market analysis.

- Help create VIX-based features to represent market uncertainty before the final trading hour.
- Suggest suitable exploratory visualisations for final-hour SPX return analysis.
- Help create a distribution plot for the final-hour SPX return.
- Help create a target class balance plot for Down, Neutral, and Up classes.
- Suggest how to visualise the largest positive and negative final-hour SPX moves.
- Help create a weekday comparison plot for average final-hour SPX returns.
- Help create a correlation matrix for engineered intraday features.
- Explain how to interpret modest correlations in a short-horizon financial forecasting problem.
- Help reduce overly numerical EDA interpretation into concise report-style wording.
- Suggest how to link EDA findings to the dissertation research questions.
- Explain why direction and magnitude both matter for 0DTE options trading evaluation.
- Help identify which EDA outputs are most suitable to include as report figures.
- Assist with exporting Python plots as PNG files for inclusion in Overleaf.
- Help troubleshoot matplotlib figure sizing for report-quality outputs.
- Suggest a clear file and folder structure for storing code, data outputs, figures, and artefact documentation.
- How can I document coding progress clearly for a dissertation artefact?
- Suggest best practices for making the notebook workflow reproducible.
- Help debug Python package and notebook execution errors during artefact development.
- Suggest baseline ML models suitable for a three-class financial prediction problem.
- Explain suitable evaluation metrics for regression models predicting return magnitude.
- Suggest how to compare model performance against simple baseline models.

- How can I convert model outputs into call, put, or no-trade signals for a simplified 0DTE strategy?
- Explain how transaction costs, bid–ask spread, option premium, and execution timing should be considered in a trading simulation.
- Suggest limitations to document when using Yahoo Finance data for an interim proof-of-concept dataset.
- Help create a weekly plan for completing the interim report, artefact, poster, and appendix documentation.
- Help break the dissertation work into short-term tasks leading up to the interim submission.
- Suggest a time management plan for balancing coding, EDA, report writing, and poster preparation.
- Help create a checklist for finalising figures, captions, references, appendices, and Overleaf formatting.
- Suggest how to prioritise remaining dissertation tasks after the interim submission.

B Github Project Link

Github link to the project including all code/reports:
<https://github.com/Ferreter/Dissertation>

C Mahara Documentation Link

Mahara Documentation Link with meeting minutes and progress:
<https://mahara.dkit.ie/view/view.php?t=a3108f281f2b65a72a1b>

D Ethical Approval Documentation

This appendix contains the ethical approval documentation completed for the dissertation project. The project does not involve human participants, vulnerable groups, surveys, interviews, focus groups, or animal research. The research uses secondary financial market data and therefore does not collect personal or identifiable information.



DUNDALK INSTITUTE OF TECHNOLOGY
School of Informatics & Creative Arts
Ethical Approval Form for Research Projects

Researcher Name: Harsh Khatri Year: 2026 Course MSC in Data Analytics

Title of project : Predicting Final-Hour SPX Price Movements for Short-Dated Options Trading Using Intraday Market Features

Name of supervisor/s Zohaib Ijaz
(if applicable)

Date 25:03:26

This application is to be completed by the researcher and where appropriate, in conjunction with the project supervisor. The lead researcher/supervisor is responsible for submitting the completed form to the appropriate Research Ethics Committee (details below)

Please note: If your submission is incomplete or unclear, your application will be returned to you, and your project may be delayed.

Section 1

Type of Researcher

Please tick the appropriate box below to indicate the type of researcher you are:

Undergraduate
(proceed to section 2)

☐

Completed Ethical Approval forms for undergraduate research should be submitted to the relevant Departmental Research Ethics Committee (DREC).

Drec.dcam@dkit.ie – Department of Creative Arts Media and Music
Drec.dcs@dkit.ie – Department of Computing Science and Mathematics
Drec.dvhcc@dkit.ie – Department of Visual and Human Centred Computing

Postgraduate
(proceed to section 3)

☒

Completed Ethical Approval forms for Postgraduate research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

Staff
(proceed to section 3)

☐

Completed Ethical Approval forms for Staff research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

ICA Research Ethics Approval Application

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Section 2

Please complete questions 1-4 listed below.

	Human and / or Animal Research	YES	NO
1	Does your research involve human participants other than the following¹? • Research using exclusively secondary sources. • Research using materials legally accessible to the public that have legal protection, e.g., record of court judgements, data archives. • Research using materials that are publicly accessible and where there is no reasonable expectation for privacy, e.g., books, published third party interviews. • Observations of human behaviour in public where (i) those being observed have no reasonable expectation of privacy, (ii) there is no intervention on the part of the researcher nor any interaction between the researcher and those observed, and (iii) individuals are not identifiable in the results. If 'YES', please complete B and C below.		
2	Does your project involve working with animals? – If 'YES', please complete B and C below. – Please note that for ethical consideration: 'Animals' are classed as vertebrate animals including cyclostomes and cephalopods (DIRECTIVE 2010/63/EU)		
3	Does your project involve working with participants from any of the following categories? • Minors (under 18 years of age) • People with learning or communication difficulties • Patients • People in custody • People engaged in illegal activities If 'YES', please complete D below.		
4	Does your project have any possible ethical implications other than those outlined in questions 1, 2 and 3? If 'YES', please complete E below.		

A. I consider that this project has no significant ethical implications² to be brought through the ICA School Ethics Review Process
Please complete Section 4

☐

¹ If there is any doubt, researchers should contact the Chair of the SREC.

² In determining significant implications, please consider all potential risks attached to this project. If there is any doubt, researchers should contact the Chair of the SREC.

- B.**
I. Is this study part of a larger project that already has ethical clearance?

YES / NO

If **YES** please answer question B II.
If **NO** please answer question C.

- II.** If this study is part of a larger project that already has ethical clearance, are you proposing any changes to the operational plan already ethically approved?

YES / NO

If **YES**, please complete *Sections 3 and 4*.
If **NO**, then please provide the project details below and complete *Section 4*.

Title of project with ethical clearance: _____

- C. Could this project have ethical implications that should be brought before the appropriate ICA Departmental Ethics Review Committee as it will be carried out with human participants?**

YES / NO

If **YES**, please complete *Sections 3 and 4*.
If **NO**, please complete *Section 4*.

- D. I consider that this project may have ethical implications that should be brought before the School Research Ethics Committee as it will be carried out with human participants in a "vulnerable" category.** ☐
Please complete Sections 3 and 4

All research carried out with human participants in a vulnerable category must be referred by the Departmental Research Ethics Committee to the School Research Ethics Committee for approval.

- E. Could this project have ethical implications, other than those previously outlined, that should be brought before the appropriate ICA Departmental Ethics Review Committee?**

YES / NO

If **YES**, please complete *Sections 3 and 4*.
If **NO**, please complete *Section 4*.

Section 3

3.1 Application Form Checklist

Please complete Section 3 and provide additional information as attachments.

My application includes the following documentation:	INCLUDED (mark as YES)	NOT APPLICABLE (mark as N/A)
Recruitment advertisement		NA
Participant Information Leaflet		NA
Participant Informed Consent form		NA
Questionnaire/Survey		NA
Interview/Focus Group Questions		NA
Debriefing material		NA
Evidence of approval to gain access to off-site location		NA
Ethical approval from external organizations.		NA
If ethical approval from external organizations is pending give details below		
Details		

3.2 Project Details

a) Lay description (Maximum 200 words)

Please outline, in terms that any non-expert would understand, what your research project is about, including what participants will be required to do. Please explain any technical terms or discipline-specific phrases.

In the last hour of trading, financial markets frequently see increased volatility, especially in the S&P 500 index (SPX), where short-dated options are extremely vulnerable to sudden price swings. Although it is challenging to consistently predict these short-term movements, they can present trading opportunities. This dissertation explores the possibility of using intraday market behavior patterns to forecast notable price changes during the last trading hour. Machine learning models will be created to determine whether the market is likely to move significantly upward, downward, or remain relatively stable prior to market close by analyzing historical price and volatility data. Additionally, the study will assess whether these forecasts can enhance decision-making in a streamlined end-of-day options trading approach. The objective is to close the gap between statistical modeling and real-world trading applications.

b) Research objectives (Maximum 150 words)

Please summarise briefly the objectives of the research.

RQ1:
Can intraday market features be used to predict the direction of SPX price movements during the final hour of trading?
RQ2:
To what extent can intraday features predict the magnitude of SPX price movements during the final hour of trading?
RQ3:
Which intraday features (e.g., momentum, volatility, and price positioning) are most important in predicting large end-of-day movements?
RQ4:
Can SPX price movements in the final hour be predicted with sufficient accuracy to be economically meaningful for trading applications?
RQ5:
Can these predictions be used to construct a profitable end-of-day options trading strategy, after accounting for transaction costs and realistic execution constraints?

c) Research location and duration

Location(s)/Population*	Dundalk Institute of Technology
Research start date	2 nd March 2026
Research end date	September 2026
Approximate duration	6 months

* If location/Population other than DkIT campus/population, provide details of the approval to gain access to that location/population as an appendix.

3.3 Participants

		YES	NO	N/A
Do participants fall into any of the following special groups?	Minors (under 18 years of age)			NA
	People with learning or communication difficulties			NA
	Patients			NA
	People in custody			NA
	People engaged in illegal activities (e.g. drug-taking)			NA
Have you given due consideration to the need for satisfactory Garda clearance?				NA

3.4 Sample Details

Approximate number	NA
Where will participants be recruited from?	NA
Inclusion Criteria	NA
Exclusion Criteria	NA
Will participants be remunerated, and if so in what form? NA	

Justification for proposed sample size and for selecting a specific gender, age, or any other group if this is done in your research. NA

3.5 Risk to Participants

- a) Please describe any risks to participants that may arise due to the research. Such risks could include physical stress, emotional distress, perceived coercion e.g. lecturer interviewing own students. Detail the measures and considerations you have put in place to minimize these risks
- b) What will you communicate to participants about any identified risks? Will any information be withheld from them about the research purpose or procedure? If so, please justify this decision.

3.6 Informed Consent

	YES	NO	N/A
Will you obtain active consent for participation?			NA
Will you describe the main experimental procedures to participants in advance?			NA
Will you inform the participants that their participation is voluntary and may be withdrawn at any point?			NA

If the research is observational, will you ask for their consent to being observed?			NA
With questionnaires, will you give participants the option of omitting questions they do not want to answer?			NA
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?			NA
Will the data be anonymous?			NA
Will you debrief participants at the end of their participation?			NA
Will your project involve deliberately misleading participants in any way, or will information be withheld? If you answer yes, give details and justification for doing this below.			NA

a) Please outline your approach to ensuring the confidentiality of data (that is, that the data will only be accessible to agreed upon parties and the safeguarding mechanisms you will put in place to achieve this.) You should include details on how and where the data will be stored, and who will have access to it.

Publicly accessible data and saved on DKIT OneDrive with password and MFA with Private access to Supervision Team and Researcher

b) Please outline how long the data will be retained for, if it will be destroyed and how it will be destroyed.

1. **Storage** : DKIT one Drive
2. **Access**: Private
3. **Communication** : Teams and Outlook
4. **Digital Platform Usage** : Mahara, DKIT one drive
5. **Data maintenance**: None destroyed after project

Section 4

Researcher I have read and I understand the DkIT Ethics Policy available from:
<https://www.dkit.ie/assets/uploads/documents/Research/Policies/DkIT%20Research%20Ethics%20Policy.pdf>

Signed: Harsh Khatri Print Name: Harsh Khatri Date: 25/03/26
(Researcher)

Supervisor: Applications for Ethical Approval of Undergraduate projects are forwarded to the Departmental Research Ethics Committee for approval or referral to the School Research Ethics Committee. Applications for Ethical Approval of Postgraduate and Staff projects are sent to the School Research Ethics Committee for Approval.

I have read and approved this form & information:

Signed: Zohaib Print Name: Dr. Zohaib Ijaz Date: 25/03/26
(Supervisor/Head of Department/ Research Centre Director/ Head of School)

There is an obligation on the researcher and/or supervisor to bring to the attention of the Departmental/School Research Ethics Committee(s): (a) Any issues with ethical implications not clearly covered by this form (b) Any ethical issues which may arise during the carrying out of the research; (c) Any ethically significant change made to the project after approval.

Section 5 (For office use only)

STATEMENT OF ETHICAL APPROVAL (FOR UNDERGRADUATE PROJECTS ONLY)

This project has been considered using agreed department procedures and is now:

Approved: ☐

Referred to the School Ethics Committee: ☐

Signed: _____ Print Name: _____ Date: _____
(Chair of Departmental Research Ethics Committee/Head of Department)

STATEMENT OF ETHICAL APPROVAL

This project has been considered using agreed School procedures and is now:

Approved: ☐

Rejected (further information sought): ☐

Chair of School Research Ethics Committee

This project has been considered by the Ethics Committee and ethical approval is granted.

Signed: _____ Print Name: _____ Date: _____
Chair of School Research Ethics Committee